

ROOPAM VERMA

Data Engineer | MSCS Graduate

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CAREER OBJECTIVE

AI-focused Software Engineer with a strong foundation in data engineering and machine learning, experienced in building scalable data pipelines and intelligent applications using Python, Spark, and cloud platforms. Hands-on with ML models, LLM APIs, agent workflows, and RAG. Seeking to contribute to production-grade AI systems.

EDUCATION

M.S. Computer Science, University of Texas, Arlington Aug 2021 – Dec 2023

Database Systems, Big Data, ML, Cloud Computing, AI, Data Mining, OS, Software Engineering

B.E. Electronics and Communication Engineering, Banasthali Vidyapith, India July 2014 – May 2018

SKILLS

Programming: Python, SQL, Scala, C++

AI & ML: Machine Learning, Deep Learning, NLP, Prompt Engineering, LLM APIs (OpenAI, Hugging Face)

AI Frameworks: LangChain, Vector Databases, RAG, Agent Workflows

Data Engineering: ETL/ELT Pipelines, Apache Spark, Databricks, Data Modeling

Cloud & DB: AWS (S3, Lambda, Redshift), Azure (ADF, Synapse), GCP (BigQuery, Dataflow), MySQL, Oracle, MongoDB

Tools: Git, CI/CD, Jupyter, Power BI, VS Code

WORK EXPERIENCE

Expert Contributor (Contract), Snorkel AI

May 2026 – Present | Remote

- Authored and validated graduate-level problems to evaluate and improve LLM reasoning, analytical depth, and solution accuracy.
- Contributed to proprietary datasets for AI fine-tuning, evaluation, benchmarking, and RLHF workflows.
- Evaluated AI-generated responses for logical consistency, mathematical correctness, and academic rigor.
- Identified reasoning gaps in deductive logic, quantitative analysis, and multi-step problem solving to improve model robustness.
- Developed automated multi-sheet financial analysis templates (PVM, Forecast Accuracy, Margin) using advanced Excel to improve FP&A; reporting efficiency.
- Supported advancement of AI reasoning benchmarks influencing next-generation model evaluation and fine-tuning strategies.

Data Engineer, Hudson Advisors

May 2023 – September 2024

- Designed and supported scalable ETL/ELT pipelines using Oracle SQL, Informatica, Databricks, and Apache Spark (Scala/Spark SQL) for enterprise-scale datasets.
- Optimized Spark jobs, reducing pipeline runtime by ~30% and improving cluster resource utilization.
- Led migration of on-premises data sources to Azure and GCP using Databricks for scalable processing.
- Implemented data quality checks, validation logic, and error handling within ETL workflows to ensure accuracy and completeness.
- Developed data models (fact and dimension tables) supporting downstream reporting, BI, and analytics.
- Collaborated cross-functionally to translate business requirements into technical specs with clear pipeline documentation.

Systems Engineer, Infosys Pvt. Ltd., India

Oct 2018 – Feb 2021

- Developed and improved the Media Gateway Control Function product (GSX) for Ribbon Communications, USA.
- Deep expertise in C++, SIP stack, SDP message parsing, call flows (SIP/ISDN), and network protocols (TCP/UDP, RTP, DNS).
- Awarded 'Winner of Hackathon' for building an audit tool to eliminate Memory Leakage, cutting manual effort from days to minutes.
- Proficient in Linux/Unix shell scripting, Python scripting, and SQL database management.

Research Intern, DRDO, Delhi, India

Jul 2017 – Dec 2017

- Researched Microstrip/Patch Antenna performance in Bluetooth (2.45GHz) and Wi-Fi (5GHz) bands using CST Microwave Studio simulations.

Intern Embedded Engineer, HCL Pvt. Ltd., Delhi, India

May 2016 – Jun 2016

- Built an RFID card reader using Atmel Studio and ISIS Proteus; designed and executed hardware simulations on ATMEGA 8.

TECHNICAL PROJECTS

Data Processing – Apache Spark: Python/Scala scripts for processing and presenting complex datasets.

ML – Diabetes Severity Prediction (UTA): Logistic Regression, KNN, Random Forest, SVM for diabetes severity prediction with user-facing webpage. Accuracy ~84%.

Student Discussion Board – Flask + GCP: Full-stack Flask app deployed on GCP (App Engine, CloudBuild, CloudSQL) with GitHub-based CI/CD pipeline.

Auto-Insurance Prediction (UTA): K-means and Hierarchical Clustering to predict auto insurance changes. Accuracy ~83%.

CIFAR-10 CNN Classifier (UTA): Multi-layer CNN with cross-entropy loss and SGD optimizer. Accuracy ~65%.

IMDB Sentiment Analysis – Naïve Bayes (UTA): Classifies reviews as positive/negative using prior/posterior probabilities with Laplace smoothing.

CERTIFICATIONS, CLUBS & ACHIEVEMENTS

- Databricks for Data Engineering | Explore Azure Databricks (Microsoft Learn) | Delta Live Tables – Databricks Academy
- Foundations: Data Analytics – Coursera (2022) | Python for ML & Data Science – UdeMy
- Outreach Lead – Innovacion Club, Banasthali Vidyapith | Volunteer – National Service Scheme
- Winner of Hackathon at Infosys (Memory Leak audit tool) | 1st place – Robot Race & Phycion (Mayukh 2k16)
- 1st place – Divertido-Twiddle (ICE Club) | Student Council House Leader | District Level Race winner